

# Fossil Formation Story

**Sequence the main stages by which remains or traces can become fossils over a very long time and distinguish evidence from a compressed story.**

[Open the original interactive lesson](#)

## Retrieve without guessing

Rapid burial can reduce disturbance and place remains beneath sediment.

Soft material usually changes or decays while harder parts, impressions or traces may remain.

More layers can build and compact over a very long time as minerals and rock processes preserve evidence.

Later movement or erosion may expose a fossil; Mary Anning used careful observation of real fossils as evidence about life long ago.

## Retrieve without guessing

Arrival decision: Rapid burial can reduce disturbance and place remains beneath sediment. Later movement or erosion may expose a fossil; Mary Anning used careful observation of real fossils as evidence about life long ago.

## How can a sequence explain fossil formation without pretending every fossil forms in exactly the same way?

Picture first: the soft parts have gone; the hard parts are being covered. Which step comes next?

How can a sequence explain fossil formation without pretending every fossil forms in exactly the same way? Soft material usually changes or decays while harder parts, impressions or traces may remain.

Checking questions:

- Rapid burial can reduce disturbance and place remains beneath sediment.
- Soft material usually changes or decays while harder parts, impressions or traces may remain.

## How can a sequence explain fossil formation without pretending every fossil forms in exactly the same way?

- More layers can build and compact over a very long time as minerals and rock processes preserve evidence.
- Later movement or erosion may expose a fossil; Mary Anning used careful observation of real fossils as evidence about life long ago.

## How can a sequence explain fossil formation without pretending every fossil forms in exactly the same way?

**2 · soft parts decay; hard parts remain; sediment settles on top**



Fossil formation stage 2: hard parts covered by sediment

# A fossil-formation sequence

## Fossil formation sequencer

Choose the five steps in order. Then answer the honesty question.

## A fossil-formation sequence

- A. more layers choose down and harden into rock
  - B. a living thing dies and sinks to the bottom
  - C. uplift or erosion brings the fossil to the surface
  - D. soft parts decay; hard parts are covered by sediment
  - E. minerals replace or fill the remains
  - F. Every fossil forms exactly this way
  - G. This is one common route; footprints, amber and moulds form differently
- A sequence is a model. Say what it does not cover.



## A fossil-formation sequence

REMAINS parts or traces left by a living thing

SEDIMENT loose material that can settle in layers

FOSSIL preserved remains, impression or trace from long ago

Order five supplied picture cards: living thing or trace; burial by sediment; change while layers build; preserved fossil in rock; later exposure. Use 'can' and 'may' because the sequence is a model, not a promise for every organism. Mary Anning is named as a public figure connected with careful fossil finding and study, not as a fictional learner or exemplar.

- Rapid burial can reduce disturbance and place remains beneath sediment.

## A fossil-formation sequence

- Soft material usually changes or decays while harder parts, impressions or traces may remain.
- More layers can build and compact over a very long time as minerals and rock processes preserve evidence.
- Later movement or erosion may expose a fossil; Mary Anning used careful observation of real fossils as evidence about life long ago.
- Sequence the main stages by which remains or traces can become fossils over a very long time and distinguish evidence from a compressed story.

## How can a sequence explain fossil formation without pretending every fossil forms in exactly the same way?

More layers can build and compact over a very long time as minerals and rock processes preserve evidence. Later movement or erosion may expose a fossil; Mary Anning used careful observation of real fossils as evidence about life long ago.

## A fossil-formation sequence

### Available evidence

Rapid burial can reduce disturbance and place remains beneath sediment.

### Reasoning move

Soft material usually changes or decays while harder parts, impressions or traces may remain.

### Boundary or next check

Later movement or erosion may expose a fossil; Mary Anning used careful observation of real fossils as evidence about life long ago.

## A fossil-formation sequence

Rapid burial can reduce disturbance and place remains beneath sediment. Later movement or erosion may expose a fossil; Mary Anning used careful observation of real fossils as evidence about life long ago.

## A fossil-formation sequence decision lab

### Supported

Order five picture cards and match one prepared explanation to each stage.

### Standard

Order and explain the stages using burial, sediment, change, preservation and exposure.

### Stretch

Explain why fossil formation is uncommon and evaluate where the simple sequence has limits.

## A fossil-formation sequence decision lab

Build a fossil-formation strip from the supplied cards. Add a short explanation beneath each stage, include 'over a very long time', and place one caution card showing that many remains never become fossils. Finish by identifying which stage is inferred from the model and which clue could be observed in a fossil or rock layer.

## Independent record · A fossil-formation sequence

Build a fossil-formation strip from the supplied cards. Add a short explanation beneath each stage, include 'over a very long time', and place one caution card showing that many remains never become fossils. Finish by identifying which stage is inferred from the model and which clue could be observed in a fossil or rock layer.

### Supported

Order five picture cards and match one prepared explanation to each stage.

### Standard



## Independent record · A fossil-formation sequence

Order and explain the stages using burial, sediment, change, preservation and exposure.

### Stretch

Explain why fossil formation is uncommon and evaluate where the simple sequence has limits.

### Trace

Rapid burial can reduce disturbance and place remains beneath sediment.

### Demand

## Independent record · A fossil-formation sequence

More layers can build and compact over a very long time as minerals and rock processes preserve evidence.

### Truth

Later movement or erosion may expose a fossil; Mary Anning used careful observation of real fossils as evidence about life long ago.

Exit route remains available: pause, use the print pack, reduce visual density or direct an adult. The named learning demand remains available.

## Audience receives; Influence changes one next action

Point to one transition in the sequence and explain what changes. The adult repeats the stage without making it certain for every fossil. Confirm, correct or add the caution card.

Final check: no result, quote, date, event or pupil wording has been created to make the map look complete. The HTML stores no learner response.